

Business Problem

The online food delivery industry handles thousands of customer orders every day. Each transaction generates valuable data related to food items, customer membership, payment methods, delivery status, and order quantity. However, when this information is stored in raw datasets, it becomes difficult for business managers to monitor performance and make timely decisions.

The goal of this analysis is to:

- Identify the most and least ordered food items.
- Analyze monthly ordering trends to understand customer demand.
- Compare ordering behavior between Gold and Regular members.
- Evaluate delivery performance by analyzing delivered and cancelled orders.
- Identify customers' preferred payment methods.
- Monitor key business KPIs for operational performance.
- Support management with data-driven decisions to improve customer satisfaction and business growth.

Dataset Overview

The dataset contains transactional data from a food delivery platform and captures customer Details, Food_Details, Resturant_Details, and Sales Data.

Dataset includes:

- customer Details (Customer_id, Customer_name, member_type)
- Food_Details (Food items, Customer Details, Resturant_Details)
- Resturant_Details (Resturant_Code, Resturant_Name, Resturant_Type)

Dashboard KPIs

- **Total Quantity:** 15K
- **Total Transactions:** 3K

- **Average Quantity per Transaction:** 5.50
- **Monthly Order Trend**
- **Food-wise Quantity Analysis**
- **Payment Method Distribution**
- **Delivery Status Analysis**
- **Member Type Analysis**

Summary Statistics

Key Business Metrics

Metric	Value	Business Insight
Total Quantity	15K	Indicates overall food demand.
Total Transactions	3K	Represents total customer orders.
Average Quantity	5.50	Customers purchase around 5–6 items per order.
Delivery Success Rate	94.25%	High operational efficiency.
Cancellation Rate	5.75%	Small percentage of cancelled orders requiring monitoring.

Order Trend

- Monthly order quantity shows changing customer demand.
- Certain months record higher order volumes, indicating seasonal demand patterns.

Food Performance

- Samosa is the highest-selling food item.
- Some food items receive comparatively fewer orders, indicating opportunities for promotional campaigns.

Payment Preferences

- UPI is the most preferred payment method.
- Card and Cash on Delivery are less frequently used.

Customer Membership

- Gold Members contribute more transactions than Regular Members, suggesting stronger customer loyalty.

Data Validation

To ensure reliable reporting and accurate business insights, the following validation checks were performed:

- Removed duplicate transaction records.
- Checked for missing or null values.
- Verified that Quantity values were greater than zero.
- Standardized Food Category and Payment Method names.
- Validated Delivery Status values (Delivered or Cancelled).
- Verified unique Transaction IDs.
- Ensured consistent Month formatting for trend analysis.
- Cross-checked KPI calculations with the original dataset.

These validation steps improved data quality and ensured that dashboard insights accurately reflect business performance.

Research Questions & Key Findings

1. Which food items are most popular?

Finding

Samosa is the highest-selling food item, while a few menu items receive comparatively lower demand.

Business Impact

- Maintain sufficient inventory for popular items.

- Promote low-selling items through discounts or combo offers.

2. How does customer demand change over time?

Finding

Monthly order quantities fluctuate, indicating seasonal demand patterns.

Business Impact

- Improve inventory planning.
- Schedule marketing campaigns during low-demand periods.

3. Which payment method do customers prefer?

Finding

UPI is the most frequently used payment method.

Business Impact

- Expand digital payment promotions.
- Offer cashback and loyalty rewards for online payments.

4. How efficient is the delivery process?

Finding

Approximately **94.25%** of orders are successfully delivered, while **5.75%** are cancelled.

Business Impact

- High delivery success reflects strong operational performance.
- Reducing cancellations can improve customer satisfaction and increase repeat orders.

5. Which customer segment contributes more?

Finding

Gold Members place more orders than Regular Members across multiple food categories.

Business Impact

- Strengthen loyalty programs.

- Encourage Regular Members to upgrade through exclusive benefits and personalized offers.

6. What are the overall business insights?

Key Findings

- Food demand is concentrated among a few popular menu items.
- Customer ordering behavior varies across months.
- Digital payments dominate customer transactions.
- Delivery operations perform efficiently with a high success rate.
- Loyal customers contribute significantly to overall business performance.